

Theoretical notes: PBHs

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1 Background

Primordial black holes are any type of black hole which form any time before big bang nucleosynthesis (not sure if this is the precise cut-off. Allowed up until slightly after universe become matter dominated). During these times there were no stars so it is not possible for BHs to form via the normal mechanism of stellar collapse. So other novel mechanisms are needed:

2 PBH formation

While black holes could, in principle (?) be formed before inflation, they density of any objects before inflation are exponentially small so we do not think of PBHs of having being from before inflation. We do not know what caused inflation, the current idea is that of a scalar field known as the inflaton. Any perturbations or fluctuations of this inflaton field cause inhomogeneities which can collapse and form black holes. The probability of fluctuations collapsing to form PBHs

In the radiation dominated universe (before inflation ends) PBH mass scales with time (heavier PBHs at later time). Q: Why? A: Unknown

2.1 PBHs as DM

PBHs are considered non-baryonic dark matter. Q: Why? A: we consider baryonic matter to be anything formed by big bang nucleosynthesis (BBN), and inferences on BBN are used to constrain the baryonic abundance. How-

ever, PBHs are expected to form before BBN, so are considered non-baryonic DM.

2.2 PBH detection

Q: How to differentiate PBHs from stellar BHs? A: PBHs can form at any mass unlike stellar BHs, so they are identifiable:

- If they are light ($\lesssim 3M_\odot$)
- If they are SMBH ($> 10^5 M_\odot$) and in the early universe ($z > 6$)
- Other more complicated techniques exist, see: Sasaki et al., 2016; Ali-Haïmoud et al., 2017; Clesse and García-Bellido, 2017; Clesse and García-Bellido, 2018; Garcia-Bellido et al., 2020; De Luca et al., 2020b; De Luca et al., 2021; Wong et al., 2021.

3 Resampling

The frequency evolution of the PBH binary is (note that we are neglecting Doppler and sidereal modulation at the moment):

$$f(\beta, t) = f_0 \left(1 - \frac{8}{3}\beta t\right)^{-3/8}, \quad (1)$$

where we have introduced the parameter:

$$\beta = \frac{96}{5}\pi^{8/3} \left(\frac{G}{c^3}\right)^{5/3} M_c^{5/3} f_0^{8/3}. \quad (2)$$

Fig. 1 shows the values of beta for the parameter space of interest. The corresponding phase (aside from a constant term) is:

$$\phi(\beta, t) = -\frac{6\pi}{5} f_0 \frac{(1 - \frac{8}{3}\beta t)^{5/8}}{\beta}. \quad (3)$$

Our ultimate aim is to be able to write the phase as:

$$\phi(\beta, \tau) = 2\pi f_0 \tau(\beta), \quad (4)$$

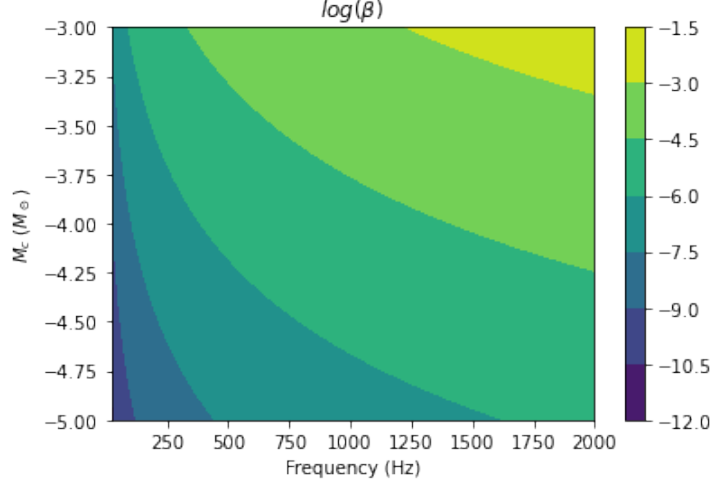


Figure 1: Range of β for the parameter space of interest. β spans approximately 10 orders of magnitude.

since that would ensure that all the signal power is confined to a single frequency bin when we Fourier transform. This can be achieved by redefining the time as:

$$\tau(\beta) = -\frac{3}{5} \frac{(1 - \frac{8}{3}\beta t)^{5/8}}{\beta} \quad (5)$$

This redefinition of time is also known as stroboscopic resampling (22/01/24: this is not true) and there are some extra details at [3, 1].

Since our data $d(t)$ is a discrete time series sampled uniformly in our original time parameter, we need to construct a new time series $d'(t')$ which would be our new data time series sampled in our new time variable. To do this we use nearest neighbour interpolation, which just defines our new time series by $d'(t') = d(t \approx t)$, i.e. each point in the new time series takes the value of the original time series at the original time t closest to our redefined time t' (for each value in t'). It is also possible to use a method of interpolation detailed in Ref. [1] which calculates the new time series by padding the Fourier series in the frequency domain before transforming back to the time domain.

The need for this interpolation, either nearest neighbour or otherwise, explains why the original time series needs to have a higher sampling frequency,

as this reduces the interpolation error. The resampled time series needs only to satisfy the Nyquist sampling theorem. The sampling frequency ratio between original and resampled data needs to be approximately 10 times bigger, but the exact condition on this ratio is not well understood and needs to be studied more in detail. Since the necessary sampling frequency of both original and resampled times series are determined by the sampling theorem, this motivates why it is preferential to study the early part of the inspiral, which has a lower frequency, meaning a lower sampling frequency is necessary and the analysis is computationally cheaper.

In the case of real data where the value of β is not known a priori, the algorithm would require the construction of a grid in β (described in next section). For each point on the grid, stroboscopic resampling is performed and the resulting power spectrum is analysed. The value of β with the strongest peaks in the power spectrum should correspond to the true value of β . The location of the peaks in the power spectrum give the value of f_0 , given the pair (f_0, β) the chirp mass can be calculated from the definition of β in Eq. 2.

3.1 Some useful approximations and rewritings

β can also be rewritten (exactly) as:

$$\beta = 3.7 \times 10^{-4} \left(\frac{M_c}{10^{-2} M_\odot} \right)^{5/3} \left(\frac{f_0}{200 Hz} \right)^{8/3} \quad (6)$$

The time taken for the signal to go from f_0 to f_{max} can be written as:

$$t_{max} = \frac{0.375}{\beta} \left(1 - \left(\frac{f_0}{f_{max}} \right)^{8/3} \right) \quad (7)$$

$$= 1013.75 \left(\frac{M_c}{10^{-2} M_\odot} \right) \left(\left(\frac{f_0}{200 Hz} \right)^{-8/3} - \left(\frac{f_{max}}{200 Hz} \right)^{-8/3} \right) \quad (8)$$

4 How to implement resampling

The Rome group has an implementation of resampling known as **strobo** which implements stroboscopic resampling as described in [3, 1]. The fundamental difficulty is that the data is initially uniformly sampled in time.

However, this means that the data will not be uniformly sampled in the resampled time. This is a problem because the standard FFT algorithm requires uniformly sampled data.

strobo solves this by using nearest-neighbour interpolation. The initial time-series is taken at a higher sampling rate and the resampled (uniformly sampled) time-series is calculated by considering taking the value at the nearest (I think the actual implementation is taking the value at the next time point) sampled time point.

I have created a new implementation which uses a non-uniform fast fourier transform (NUFFT). While the classic FFT implementation requires uniformly sampled data, there is a range of literature determining the optimal approximation techniques for calculating the fft for non-uniform data. See [?, ?, ?] for and reference within for a range of different approximation techniques. I use the python implementation described in [?] described in [?, ?, ?]. However, a large number of other python implementations exist, see [?, ?, ?], which have not been rigorously benchmarked against. See also [?] for an implementation in matlab.

My nufft implementation on the CPU (using the finufft package) is 10x faster than the strobo implementation. This is primarily because we can use a lower sampling rate for the original time series. This was previously required to ensure the errors of the nearest-neighbour interpolation was small.

5 Constructing the grid on β

We want to determine the spacing of the grid on β (i.e. the step $\Delta\beta$) which still ensures the the phase difference between 2 signals characterised by two values β and $\beta + \Delta\beta$ is smaller than some threshold $\Delta\phi_{thr}$.

Let us first consider the analytic expressions for the dephasing between $\phi_{exact} = \phi(\beta)$ and $\phi_{offset} = \phi(\beta + \Delta\beta)$. There is a constant dephasing

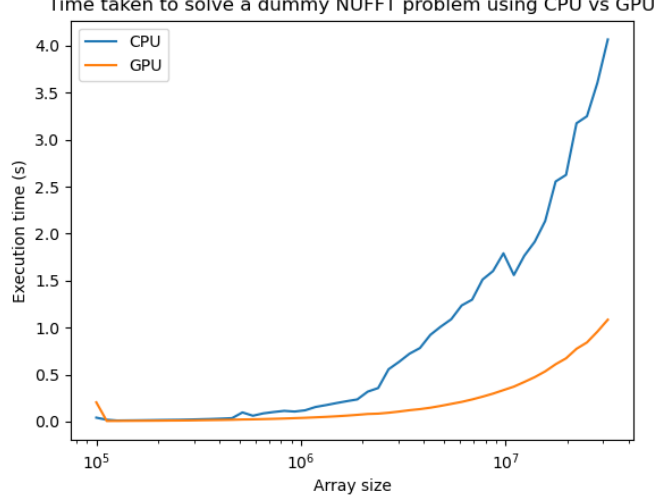


Figure 2: Timing performance of CPU vs GPU implementation of the NUFFT on a dummy problem

term at $t = 0$ which is given by:

$$\Delta\phi(t=0) = \phi_{offset}(t=0) - \phi_{exact}(t=0) \quad (9)$$

$$= -\frac{6\pi f_0}{5} \left(\frac{1}{\beta + \Delta\beta} - \frac{1}{\beta} \right) \quad (10)$$

$$= \frac{6\pi f_0}{5} \frac{\Delta\beta}{\beta^2 + \beta\Delta\beta} \quad (11)$$

Working with the exact expression for ϕ_{offset} isn't useful to find a expression for the maximum possible $\Delta\beta$ so we instead Taylor expand for small $\Delta\beta$:

$$\phi_{approx}(t) = T.E_{\Delta\beta} [\phi_{offset}(t)] \quad (12)$$

We use this to calculate the dephasing:

$$\Delta\phi_{approx}(t) = \phi_{approx}(t) - \phi_{exact}(t) \quad (13)$$

$$= \frac{6\pi f_0}{5} \frac{\Delta\beta}{\beta} \frac{-1 + \beta t}{\beta(1 - \frac{8}{3}\beta t)^{3/8}} \quad (14)$$

We can also write this w.r.t the resampled time:

$$= 2\pi f_0 \tau \frac{\Delta\beta}{\beta} \frac{3^{8/5} 5^{-3/5} + 3(\beta\tau)^{8/5}}{8(\beta\tau)^{8/5}} \quad (15)$$

Note, however, that this approximation incorrectly gives the constant dephasing term:

$$\frac{\Delta\phi_{approx}(t=0)}{\Delta\phi(t=0)} = \frac{\Delta\beta}{\beta^2} / \frac{\Delta\beta}{\beta^2 + \beta\Delta\beta} \quad (16)$$

$$= 1 + \frac{\Delta\beta}{\beta} \quad (17)$$

This is because we are around small $\Delta\beta$ and not w.r.t t so each term will have a contribution to the $\Delta\phi(t=0)$.

In reality this constant phase term doesn't matter for the frequency evolution. Therefore we introduce a constant to set the $\Delta\phi_{approx}(t=0) = 0$:

$$\Delta\phi_{approx}(t) = \frac{6\pi f_0}{5} \frac{\Delta\beta}{\beta} \frac{-1 + \beta t}{\beta(1 - \frac{8}{3}\beta t)^{3/8}} + \frac{6\pi f_0}{5} \frac{\Delta\beta}{\beta^2} \quad (18)$$

For completeness, we check that our analysis is valid by making sure the 2nd order term in the Taylor Expansion is small over the space we care about:

$$\text{2nd order term} \propto \frac{\beta^2 \Delta\beta^2 t^2}{\beta^{35/8}} \quad (19)$$

$$= \frac{\epsilon^2 t^2}{\beta^{3/8}} \quad (20)$$

where $\epsilon = \frac{\delta\beta}{\beta}$

$$(21)$$

We can now compute the maximum permitted $\Delta\beta$ to ensure a dephasing of no more than $\Delta\phi_{thresh}$

$$\frac{6\pi f_0}{5} \frac{\Delta\beta}{\beta} \frac{-1 + \beta t}{\beta(1 - \frac{8}{3}\beta t)^{3/8}} + \frac{6\pi f_0}{5} \frac{\Delta\beta}{\beta^2} \leq \Delta\phi_{thresh} \quad (22)$$

$$\frac{\Delta\beta}{\beta^2} \leq \frac{5\Delta\phi_{thresh}}{6\pi f_0} \left(\frac{(-1 + \beta t_{max})}{(1 - \frac{8}{3}\beta t_{max})^{3/8}} + 1 \right)^{-1} \quad (23)$$

We can rewrite this using Eq. 7:

$$\frac{\Delta\beta}{\beta^2} \leq \frac{5\Delta\phi_{thresh}}{6\pi f_0} \left(\frac{-\frac{5}{8} - \frac{3}{8} \left(\frac{f_0}{f_{max}} \right)^{8/3}}{\frac{f_0}{f_{max}}} + 1 \right)^{-1} \quad (24)$$

$$\Delta\beta \leq \frac{5\beta^2\Delta\phi_{thresh}}{6\pi f_0} \left(\frac{-\frac{5}{8} - \frac{3}{8} \left(\frac{f_0}{f_{max}} \right)^{8/3}}{\frac{f_0}{f_{max}}} + 1 \right)^{-1} \quad (25)$$

Todo list:

- Heatmap of 2nd order correction
- Dephasing into power loss
- Compare with injection studies

6 Numerical tests

6.1 Without error

We simulate a signal using a known f_0 and M_c (or equivalently β_{exact}). We resample the signal using a β_{resamp} which is some $\Delta\beta$ away from our known β_{exact} and consider how much power is lost as a function of $\Delta\beta$.

For different values of β_{resamp} the length of the resampled data will vary slightly, affecting where the Fourier bins are. It is important to control for this by 0-padding the data to some fixed length. See figure 3 for the effect of this.

At the moment the length of the padded data is set semi-automatically. A known prefactor is used to scaled in relation to the size of the array (controlled by T_{obs}). This is not fully automated and the prefactor needs ? to be adjusted for different chosen values of β_{exact} (key barrier is that the maximum length of resampled data is not known the first time round).

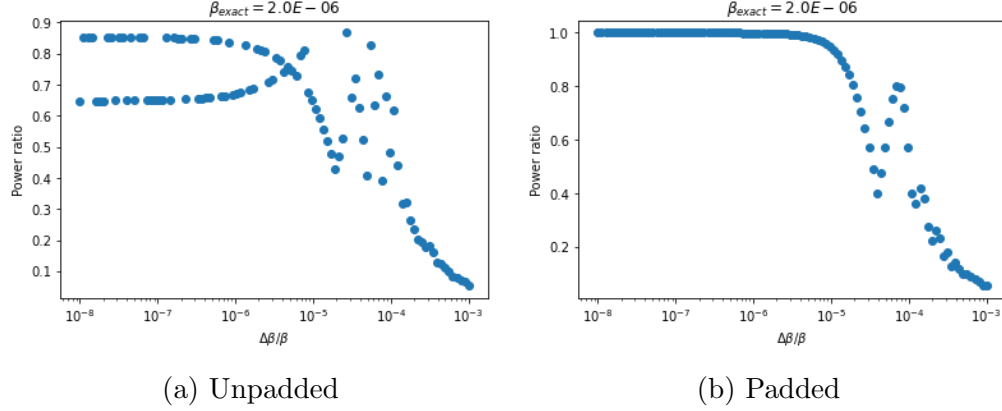


Figure 3: Power ratio calculated for different $\Delta\beta$ with padded and unpadded data.

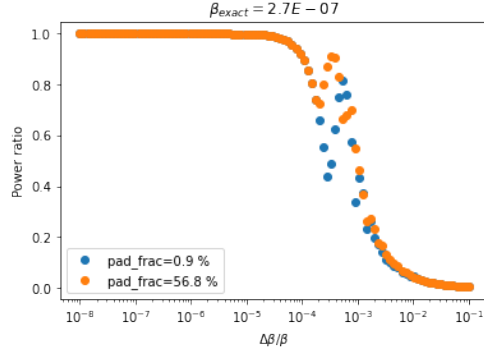
Hypothesis: the bump occurs due to the discrete nature of Fourier binning. The initial drop is caused when the center of the true Gaussian drifts close to the edge of a fft bin so the power is split between two bins. The bump occurs when the power is confined all within one fft bin again. Don't see a secondary bump because the width of the true Gaussian spreads to larger than fft bin width so where the center is doesn't matter.

Factors that affect the bump:

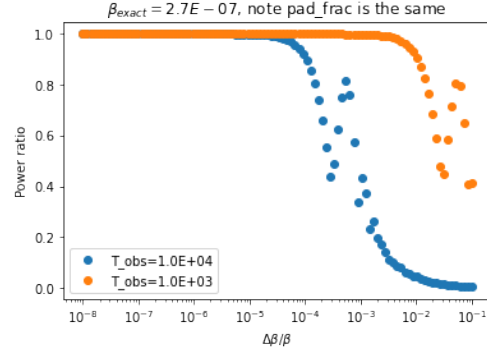
- Length of the 0-padding.
- Beta
- T_{obs}

6.2 With error

Remember that error realisations have previously affected the result. Have this working

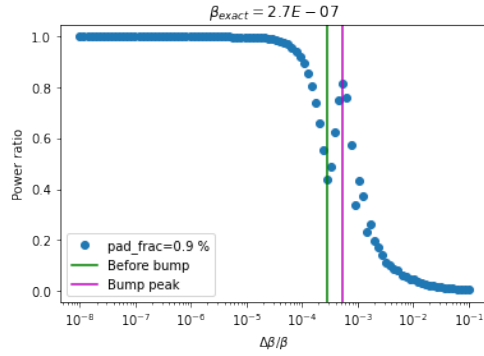


(a) Changing 0-padded length.

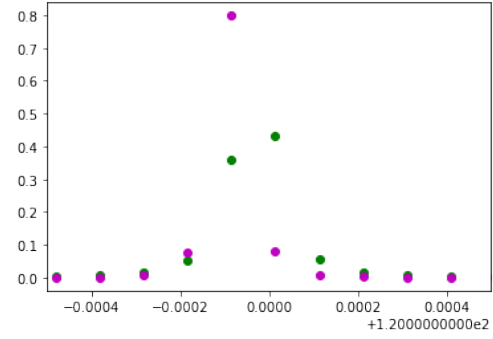


(b) Changing T_{obs}

Figure 4: Power ratio calculated for different $\Delta\beta$ with padded and unpadded data.



(a)



(b) Changing T_{obs}

Figure 5: Power spectrum before and at the peak of the bump.

6.3 Analytic distance sensitivity

6.3.1 Finding the CR^{thresh}

In this section we first consider the behaviour of real Gaussian noise $\mathcal{N}(0, \sigma)$.

The Gaussian noise was originally generated in the time domain, to compute the power spectrum $S(f)$ we need to compute the Fourier transform. The Fourier transform is complex valued (even though the noise was real valued) and the power comes from summing the squares of the real and imaginary frequency amplitudes. Recall that the Fourier transform of Gaussian noise is just a Gaussian in both the real and imaginary domain. This means that the pdf of the power spectrum is a chi-squared distribution with 2 dof, which is an exponential:

$$p(S) = \epsilon e^{-\epsilon S} \quad (26)$$

where $\epsilon = 1/\mu_s = 1/\sigma_s$ and μ_s and σ_s are the mean and standard deviation of the pdf of the noise spectrum ($\sigma_s \neq \sigma$) and for an exponential distribution we are guaranteed that $\mu_s = \sigma_s$. Note that for a particular realisation of the power spectrum, the mean and standard deviation of the realisation should also be equal to μ_s and σ_s .

Let us define the Critical Ratio as:

$$CR(f) = \frac{S(f) - \mu_s}{\sigma_s}. \quad (27)$$

The critical ratio can be interpreted as a normalisation of the power spectrum using its mean and standard deviation.

Using ϵ in the definition of critical ratio:

$$CR(f) = \epsilon S(f) - 1. \quad (28)$$

Inverting Eq. 28 for $S(CR(f))$ and substituting this into Eq. 26 (without forgetting the chain rule for pdf substitution):

$$f(CR) = e^{-CR-1} \quad (29)$$

We define the false alarm probability to be:

$$FAP = P(CR \geq CR^{thresh}) \quad (30)$$

$$= \int_{CR^{thresh}}^{\infty} e^{-CR-1} dCR \quad (31)$$

$$= e^{-CR^{thresh}-1}. \quad (32)$$

Inverting this gives:

$$CR^{thresh} = -1 - \ln(FAP) \quad (33)$$

6.3.2 Finding λ^{thresh}

We now consider the properties of a signal in the presence of noise. Instead of working with the power spectrum of this data, let us define the spectral amplitude:

$$\lambda = \frac{4|\tilde{h}(f)|^2}{TS_n(f)}, \quad (34)$$

where $\tilde{h}(f)$ is the Fourier transform of the data, $S_n(f)$ is the noise spectral density, and T is the observation time. Here we are considering real world data so $h(f)$ includes a signal and noise and $S_n(f)$ is the spectral density of the particular noise considered, e.g. of a particular detector.

Assume that the detector noise is Gaussian, in the presence of a signal you can think of the frequency spectrum of the data=signal+noise as a non-central Gaussian at the signal frequency. Since there are still real and imaginary components, the power spectrum is now calculated using a noncentral chi-squared distribution which has a probability distribution function:

$$p(S_{eq}, \lambda) = e^{-S-\lambda/2} I_0(\sqrt{2\lambda S_{eq}}). \quad (35)$$

Note, it is valid to use the analysis from the previous section since we are assuming that the detector noise is Gaussian. This assumption is commonly made for analytical calculations and verified / compared to injection studies. It is also valid within any small frequency which we approximate after resampling (because the non-Gaussian noise should be incoherent with resampling). In general, non-stationarity of the noise is more important than any non-Gaussian features (but we do not deal with it here). More details

about this should be available in LVK detector calibration papers or targetted searches (but a rigorous literature review hasn't been conducted yet).

With this in mind we substitute Eq. 28 with $\epsilon = 1$ which should be ensured by the normalisation in Eq. 34.

$$p(CR, \lambda) = e^{-CR-1-\lambda/2} I_0(\sqrt{2\lambda(CR+1)}) \quad (36)$$

Detection probability is now defined to be:

$$P_{det}(\lambda^*) = P_{\lambda}(CR \geq CR^{thresh}) = \int_{CR^*}^{\infty} p(CR, \lambda) dCR. \quad (37)$$

Inutitively, here we want to find the loudness of the signal λ^* necessary to ensure that the probability of it being detected inside a noise realisation is $P_{det}(\lambda^*)$. The concept of detection is defined by the False Alarm Probability fixed earlier, i.e. a signal is classed as a detection if the chances of it being generated by pure noise are $<$ the FAP. To achieve this, the critical ratio of a signal must be greater than the threshold critical ratio CR^{thresh} and the probability of this is exactly equation 37.

In reality, we fix a detection probability we want. We would then numerically evaluate this Eq. 37 to find the necessary threshold spectral amplitude (λ^*). Note this works because Eq. 36 depends on 2 variables: CR which is integrated over, and lambda which we are numerically solving for. In the next section we will see how to convert this numerically solved for λ^* into a distance sensitivity.

Minor note, for real data the λ and corresponding CR depend on frequency. We treat it in Eq. 37 as a scalar since we assume that the real signal frequency gives a maximum CR that is much larger than the noise. Of course any frequency with a sufficient CR is a detection which would physically mean the power of a signal is diffused into multiple FFT bins or that there is another seperate signal which is also detectable (unlikely since we are correcting for sky location) but these are both perfectly valid scenarios.

6.3.3 Calculating the distance sensitivity

Consider the case where resampling has successfully converted a long-transient signal into a purely monochromatic signal $h(t)$. From JKS Eqs. (19) - (22),

the original signal model was:

$$h(t) = h_0 F_+ A_+ \cos 2\Psi + h_0 F_x A_x \sin 2\Psi \quad (38)$$

where $F_{+/x}$ are the detector beam pattern functions and $A_+ = \frac{1+\cos^2\iota}{2}$, $A_x = \cos\iota$, note that we have set the wobble angle (in JKS) to $\theta = 90^\circ$. Prior to resampling Ψ contained terms that accounted for the proper motion of the source w.r.t the detector frame of motion but after resampling (which also removes the Doppler modulation, etc) this is now purely monochromatic.

The power of the resampled signal is then:

$$|\tilde{h}(f)|^2 = h_0^2 T^2 \frac{F_+ A_+ + F_x A_x}{4} \left(\frac{\sin \omega_k}{\omega_k} \right)^2 \quad (39)$$

where $\omega_k = \pi(f - f_k)T$, where f_k is the frequency at the center of the FFT bin. The $1/4$ and $\frac{\sin \omega_k}{\omega_k}$ factors are derived in Eqs. (B10) - (B15) of [2]. Note that Eq. (B12) is a continuous integration but for actual data and a DFFT this is a discrete sum over Fourier bins which where a delta function kills the sum and keeps only a singular bin.

We now average over the frequencies inside the Fourier window $[f_k - \delta f, f_k + \delta f]$ to calculate the average power loss as a result of the frequency mismatch. We do this by changing variables to $x = \pi(f - f_k)T$ to get:

$$\left\langle \frac{\sin \omega_k}{\omega_k} \right\rangle_f^2 = \frac{1}{\delta f} \int_{f_k - \delta f/2}^{f_k + \delta f/2} \left(\frac{\sin \omega_k}{\omega_k} \right)^2 df \quad (40)$$

$$= \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \left(\frac{\sin x}{x} \right)^2 dx = \frac{2.4308}{\pi} \quad (41)$$

We next aim to average over polarization angle ψ and inclination angle ι . From JKS Eqs. (10) - (11) the definition of the detector beam patterns are:

$$F_+ = a(t) \cos 2\psi + b(t) \sin 2\psi \quad (42)$$

$$F_x = b(t) \cos 2\psi - a(t) \sin 2\psi \quad (43)$$

where we have set $\xi = 90^\circ$ into the JKS formulae which is true for all the LVK detectors. Averaging over ψ and $\cos \iota$ gives:

$$\langle (F_+ A_+ + F_x A_x)^2 \rangle_{\psi\iota} = \frac{4}{5} \frac{a^2(t) + b^2(t)}{2} \quad (44)$$

where $a(t)$ and $b(t)$ are defined in Eqs. (12) and (13) in JKS (they are very heavy). $a(t)$ and $b(t)$ depend on the source sky location and the Earth's motion around the sun. Integrating over long times for the galactic centre sky location we numerically determine that $\langle a^2(t) + b^2(t) \rangle_t = 0.416$ for the L detector and very similar for H and V.

If we substitute Eqs. 41 and 44 into Eq. 39 we get:

$$\langle |\tilde{h}(f)|^2 \rangle_{\psi, \iota, f, t} = \frac{2.4308}{\pi} \frac{4}{5} \frac{0.416}{2} \frac{h_0^2 T^2}{4} \quad (45)$$

We insert into the definition of spectral amplitude from Eq. 34:

$$\langle \lambda(f) \rangle_{\psi, \iota, f, t} = \frac{0.40447T}{\pi} \left\langle \frac{h_0^2(f)}{S_n(f)} \right\rangle_t \quad (46)$$

We have not yet properly integrated over time since the signal amplitude also depends on time as:

$$h_0(f(t)) = \frac{4}{d} \left(\frac{GM_c}{c^2} \right)^{5/3} \left(\frac{\pi f(t)}{c} \right)^{2/3} \quad (47)$$

We substitute the expression for the frequency evolution in Eq. 1 and get:

$$\left\langle \frac{h_0^2(t)}{S_n(f(t))} \right\rangle_t = \frac{16}{d^2} \left(\frac{5}{96} \right)^2 \frac{\beta^2 c^2}{\pi^4 f_0^4 T} \int_{t_0}^{t_0+T} \frac{(1 - \frac{8}{3}\beta t)}{S_n(f(t))} dt \quad (48)$$

We substitute Eq. 48 into Eq. 46 and rearrange for distance to finally get:

$$d = \sqrt{\frac{0.02115}{\lambda^*} \frac{5c\beta}{96f_0^2}} \sqrt{\int_{t_0}^{t_0+T} \frac{(1 - \frac{8}{3}\beta t)^{-1/2}}{S_n\left(f_0 \left(1 - \frac{8}{3}\beta t\right)^{-3/8}\right)} dt} \quad (49)$$

Is there a $1/\sqrt{T}$ term???

6.4 Working abstract: A resampling search method for gravitational waves from light primordial black hole binaries

Primordial black holes are proposed to have formed in the early universe and could be valuable probes for cosmology, dark matter, and physics of

the very early universe. Sub-solar mass primordial binaries with masses of $\mathcal{O}(10^{-3}) - \mathcal{O}(10^{-1})M_{\odot}$ would generate gravitational waves with durations of hours to years, possibly detectable by ground-based detectors. I will present an improved implementation of the resampling algorithm to search for such signals. A sensitivity estimate shows that this new method can reach extra-galactic sources with current gravitational wave data, significantly increasing the horizon distance to primordial black hole binaries.

Primordial black holes are proposed to have formed in the very early universe and a first detection of them would provide valuable insights into cosmology, dark matter, and physics of the very early universe. Primordial black holes could form binaries systems with chirp masses of the order of $\mathcal{O}(10^{-5})M_{\odot} - \mathcal{O}(10^{-3})M_{\odot}$, which would emit long transient gravitational waves signals that last of the orders of hours - years. We present an implementation of a resampling algorithm to search for such signals in gravitational wave data. A sensitivity estimate shows that this new method can reach extra-galactic sources, significantly increasing the horizon distance to primordial black hole binaries.

$$[10^{-3} - 10^{-1}M_{\odot}]$$

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